

# CJ2 Voice Emergency Checklist

## Technical Data Packet

Cessna Citation CJ2 (CE 525A) — bench / training prototype

Voice-driven, fully offline emergency-checklist reader

Hardware reference, firmware behavior, processor selection,  
checklist library, and wiring diagram.

### DEMO / TRAINING USE ONLY — NOT FOR ACTUAL FLIGHT OPERATIONS

Not airworthy, not certified. Must never be installed in or relied upon aboard an aircraft.

Checklist text reproduced for demonstration/training only.

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# 1 System Overview

An offline, on-device voice-driven emergency checklist reader for the Cessna Citation CJ2 (CE 525A). The pilot states an emergency by name; the device reads each checklist item aloud from pre-recorded audio and waits for the spoken completion word before advancing. It runs entirely on an ESP32-S3 using Espressif's ESP-SR speech framework (WakeNet wake word + MultiNet offline English command recognition) — no cloud, no Wi-Fi, no phone. All GPIO numbers in this packet match `main/board_pins.h`.

| Item                 | Value                                                                                                                                                                                                                                             |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MCU                  | ESP32-S3 (dual-core LX7 @ 240 MHz) — PSRAM required by ESP-SR                                                                                                                                                                                     |
| Recommended module   | ESP32-S3-WROOM-1 N16R8 (16 MB flash, 8 MB octal PSRAM)                                                                                                                                                                                            |
| Speech stack         | Espressif ESP-SR: AFE (NS/VAD) → WakeNet “Hi ESP” → MultiNet English                                                                                                                                                                              |
| Audio input          | Galvanically-isolated audio-panel tap on I2S_NUM_0 (codec ADC / analog line tap, selectable per install). Onboard INMP441 mic = bench-test only.                                                                                                  |
| Audio output         | Isolated line driver / DAC (PCM5102A-class) on I2S_NUM_1 → output isolation transformer → line-level out to dedicated COM3-style audio-panel channel. Onboard MAX98357A speaker-amp = bench-test only; no onboard speaker in the installed build. |
| Config storage       | Two microSD slots — Config Card (slot 1, write-protected) + Data Card (slot 2), FAT32                                                                                                                                                             |
| Audio I/O posture    | Receive-only on the INPUT tap (I2SO) + galvanically-isolated OUTPUT into a dedicated COM3-style channel (I2S1). NOT a COM-radio transceiver or control/command interface.                                                                         |
| Install input source | Aircraft audio panel (analog isolated line tap OR buffered digital I2S codec ADC). INMP441 MEMS mic = bench-test development only, not the installed source.                                                                                      |
| Install output       | Isolated line-level TX to dedicated COM3-style audio-panel channel via isolation transformer. MAX98357A speaker-amp + 4–8 Ω speaker = bench-test only.                                                                                            |
| Annunciation         | Applied Avionics split-legend switch (dark-cockpit, FAA AC 25-11)                                                                                                                                                                                 |
| Logic level          | 3.3 V (ESP32-S3 is not 5 V tolerant on GPIO)                                                                                                                                                                                                      |

Two build paths are supported:

- Integrated: ESP32-S3-Korvo-2 dev board (on-board dual mic, ES8311 codec, NS4150 amp, microSD slot). Least wiring; best mic performance.
- DIY (this document): ESP32-S3 DevKitC-1 N16R8 + audio-panel input stage (isolation transformer + I2S codec ADC for analog, or buffered receive-only I2S for digital; INMP441 bench-only) + isolated audio output stage (I2S DAC e.g. PCM5102A-class + output isolation transformer → COM3-style channel; MAX98357A speaker-amp bench-only) + two microSD breakouts (Config Card slot 1 + Data Card slot 2) + the annunciator switch.

## 2 Master Pin Map

All GPIO assignments from `main/board_pins.h`.

| Function      | Macro       | GPIO | Dir     | Notes                      |
|---------------|-------------|------|---------|----------------------------|
| Push-to-talk  | PTT_GPIO    | 0    | in (PU) | BOOT button; active-low    |
| SELECT switch | SELECT_GPIO | 10   | in (PU) | IN = GPIO→GND (active-low) |

| Function              | Macro             | GPIO | Dir     | Notes                                                                                                          |
|-----------------------|-------------------|------|---------|----------------------------------------------------------------------------------------------------------------|
| Legend OFF (white)    | LEGEND_OFF_GPIO   | 21   | out     | drives top legend half (via driver)                                                                            |
| Legend FAULT (amber)  | LEGEND_FAULT_GPIO | 14   | out     | drives bottom legend half (via driver)                                                                         |
| Status LED            | STATUS_LED_GPIO   | 48   | out     | on-board RGB on most S3 devkits                                                                                |
| Audio-in bit clock    | AIN_BCLK_GPIO     | 4    | out     | I2S0 BCLK → audio-panel input codec ADC / tap (RX)                                                             |
| Audio-in word select  | AIN_LRCLK_GPIO    | 5    | out     | I2S0 WS → audio-panel input codec ADC / tap (RX)                                                               |
| Audio-in data         | AIN_DIN_GPIO      | 6    | in      | I2S0 → audio-panel input codec ADC / tap (RX)                                                                  |
| SD clock              | SD_CLK_GPIO       | 7    | out     | SDMMC CLK                                                                                                      |
| SD command            | SD_CMD_GPIO       | 9    | i/o     | SDMMC CMD (needs pull-up)                                                                                      |
| SD data 0             | SD_D0_GPIO        | 8    | i/o     | SDMMC DAT0 (needs pull-up)                                                                                     |
| Audio-out bit clock   | AOUT_BCLK_GPIO    | 15   | out     | I2S1 BCLK → isolated audio-out (COM3) line driver/DAC                                                          |
| Audio-out word select | AOUT_LRCLK_GPIO   | 16   | out     | I2S1 WS → isolated audio-out (COM3) line driver/DAC                                                            |
| Audio-out data        | AOUT_DOUT_GPIO    | 17   | out     | I2S1 DOUT → isolated audio-out (COM3) line driver/DAC → isolation transformer → COM3-style audio-panel channel |
| Config-card detect    | SD_CFG_CD_GPIO    | 47   | in (PU) | slot-1 card-detect (Config Card)                                                                               |
| Data-card detect      | SD_DAT_CD_GPIO    | 38   | in (PU) | slot-2 card-detect (Data Card)                                                                                 |

Polarity macros (flip to match your hardware): PTT\_ACTIVE\_LOW=1, SELECT\_ACTIVE\_LOW=1, LEGEND\_OFF\_ACTIVE\_HIGH=1, LEGEND\_FAULT\_ACTIVE\_HIGH=1, STATUS\_LED\_ACTIVE\_HIGH=1, LAMP\_TEST\_MS=2000. Set any unused output to -1 to disable it cleanly.

### Reserved / avoid pins (ESP32-S3-WROOM-1 N16R8)

| Pins           | Reason                                                           |
|----------------|------------------------------------------------------------------|
| GPIO33-37      | Octal PSRAM/flash internal bus — do not use on R8 modules        |
| GPIO19, GPIO20 | USB D-/D+ (native USB / USB-Serial-JTAG)                         |
| GPIO0, 45, 46  | Strapping pins — usable but must be in the correct state at boot |
| GPIO26-32      | SPI flash on some modules — verify your board                    |

GPIO0 is used here only as the BOOT/PTT button (its natural strapping use), so it is safe. None of the chosen pins conflict with the reserved set.

## 3 Per-Device Wiring

### 3.1 Audio-Panel Input Stage → ESP32-S3 (I2S\_NUM\_0)

The installed input is a galvanically-isolated tap of the aircraft audio panel, not an onboard microphone. The source is selected by the Config Card: either (a) an analog isolated line tap (600  $\Omega$  aviation audio isolation transformer, e.g. Allen Avionics AGL series) feeding an I2S codec ADC, or (b) a buffered receive-only digital I2S feed from the same codec ADC. I2S bus: AIN\_BCLK\_GPIO (GPIO4) / AIN\_LRCLK\_GPIO (GPIO5) / AIN\_DIN\_GPIO (GPIO6). This gives already-mixed crew audio from the panel, avoiding cockpit acoustic noise entirely.

- Analog path: audio-panel receive tap → 600  $\Omega$ :600  $\Omega$  isolation transformer (Allen Avionics AGL or equivalent) → I2S codec ADC → GPIO4/5/6 (I2S\_NUM\_0). The isolation transformer provides galvanic isolation; no signal path back toward the panel.
- Digital path: buffered receive-only I2S/line feed from codec ADC → GPIO4/5/6 (I2S\_NUM\_0). There is no I2S output toward the audio panel on this bus — it is receive-only.
- Bench-test only (not the installed input): INMP441 MEMS mic (1.8–3.3 V, ~2.2–2.5 mA at 3.3 V) can be wired to GPIO4/5/6 on the bench for audio-content verification before the isolated input stage is fitted. [INMP441 datasheet](#).

#### 3.1a Bench-test input: INMP441 wiring (GPIO4/5/6) — bench verification only, not the installed source

BENCH-TEST ONLY — the INMP441 mic is used for bench audio-content verification only. The installed configuration uses the isolated audio-panel tap (§3.1 above) on the same GPIO4/5/6 bus.

| INMP441 pin | Connects to           | Net                  |
|-------------|-----------------------|----------------------|
| VDD         | ESP 3V3               | 3.3 V                |
| GND         | ESP GND               | GND                  |
| SCK         | ESP GPIO4 (AIN_BCLK)  | audio-in I2S         |
| WS          | ESP GPIO5 (AIN_LRCLK) | audio-in I2S         |
| SD          | ESP GPIO6 (AIN_DIN)   | audio-in I2S         |
| L/R         | GND                   | selects left channel |

- Decouple VDD→GND with 0.1  $\mu$ F close to the module.
- Add a 100 k $\Omega$  pulldown on the SD line (datasheet-recommended to discharge the bus when tri-stated). Most breakouts include it.
- Do not clock WS/SCK with VDD unpowered (stresses ESD diodes).

### 3.2 Isolated Audio Output Stage → COM3-style audio-panel channel (I2S\_NUM\_1) — Installed Configuration

D.3.3a (installed output). Checklist read-aloud audio is sent out through a dedicated, galvanically-isolated line-level output stage on I2S\_NUM\_1 (AOUT\_BCLK\_GPIO GPIO15, AOUT\_LRCLK\_GPIO GPIO16, AOUT\_DOUT\_GPIO GPIO17). The I2S DAC (e.g. PCM5102A-class) drives a 600  $\Omega$ :600  $\Omega$  line-level isolation transformer (e.g. Allen Avionics AGL series or equivalent) on the TX output line, then line-level output to a dedicated COM3-style audio-panel input channel. The crew hears checklist items in-headset via that channel. The isolation transformer on the TX line provides a galvanic barrier; a device fault cannot key, jam, load, or back-feed the panel's other channels or required COM radios. The onboard speaker/amplifier is removed in the installed configuration.

- *Judgment call / open item:* the exact I2S DAC part and isolation transformer type must be confirmed by the bench engineer. Options include a PCM5102A-class DAC IC followed by a 600Ω:600Ω aviation audio isolation transformer. AOUT\_BCLK/LRCLK/DOOUT macros (GPIO15/16/17) share pin assignments with the former speaker-amp; the installed stage replaces the bench speaker-amp at those pins.
- Supply: a few mA for the line-level output stage (not the ~650 mA peak of the speaker-amp, which is removed). See power budget (§5) for updated 5 V rail loading.

### 3.2a Bench-test speaker output (MAX98357A) — bench verification only, not the installed output

BENCH-TEST ONLY (D.3.3b) — the MAX98357A Class-D amplifier and speaker are used for bench verification of audio content before the isolated output stage is fitted. They are not the installed output. The installed output is the isolated line driver / DAC stage (§3.2 above) feeding the COM3-style audio-panel channel.

Supply 2.5–5.5 V; 2.4 mA quiescent; peak speaker current up to ~650 mA at 5 V/4 Ω (bench figure only — not relevant to the installed output stage). No MCLK required. [Analog Devices datasheet](#), [Adafruit guide](#).

| MAX98357A pin | Connects to (bench only)                                      | Net           |
|---------------|---------------------------------------------------------------|---------------|
| VIN           | 5 V (USB/VBUS) for full output (3.3 V also works, less power) | 5 V           |
| GND           | ESP GND                                                       | GND           |
| BCLK          | ESP GPIO15 (AOUT_BCLK)                                        | audio-out I2S |
| LRC           | ESP GPIO16 (AOUT_LRCLK)                                       | audio-out I2S |
| DIN           | ESP GPIO17 (AOUT_DOOUT)                                       | audio-out I2S |
| GAIN          | NC = 9 dB (default)                                           | —             |
| SD (mode)     | float = mono (L+R)/2                                          | —             |
| OUT+/OUT-     | bench speaker (4–8 Ω) — bridge-tied, no ground ref            | —             |

- GAIN options: 100 kΩ→GND = 15 dB; →GND = 12 dB; NC = 9 dB; →VIN = 6 dB; 100 kΩ→VIN = 3 dB.
- SD/mode pin: <0.16 V = shutdown; 0.16–0.77 V = (L+R)/2 mono; 0.77–1.4 V = right; >1.4 V = left. Breakouts usually float it to mono.
- The OUT pins are bridge-tied — never connect either to GND.

### 3.3 microSD Cards → ESP32-S3 (SDMMC, 1-bit) — Two-Card Model

Two separate microSD slots: slot 1 = Config Card (write-protected per installation; mounts at /sdcard-config) and slot 2 = Data Card (checklist library + audio; mounts at /sdcard-data). Both 3.3 V, sleep ~100–200 μA; init/read peaks 50–200 mA. [SD current notes](#).

The firmware boots Config Card first: reads /sdcard-config/config.json, inits the audio source and VOX parameters, then loads the matching aircraft folder from the Data Card. Absent or unreadable Config Card = FAULT\_NO\_CONFIG; absent Data Card = FAULT\_NO\_CARD. Both slots are front-accessible.

| SD signal               | ESP32-S3                | Net         | Pull-up          |
|-------------------------|-------------------------|-------------|------------------|
| CLK                     | GPIO7                   | microSD     | —                |
| CMD                     | GPIO9                   | microSD     | 10 kΩ → 3V3      |
| DATO                    | GPIO8                   | microSD     | 10 kΩ → 3V3      |
| VDD                     | 3V3                     | 3.3 V       | —                |
| VSS                     | GND                     | GND         | —                |
| CD (slot 1 Config Card) | GPIO47 (SD_CFG_CD_GPIO) | card-detect | 10 kΩ → 3V3 (PU) |

| SD signal             | ESP32-S3                | Net         | Pull-up                  |
|-----------------------|-------------------------|-------------|--------------------------|
| CD (slot 2 Data Card) | GPIO38 (SD_DAT_CD_GPIO) | card-detect | 10 k $\Omega$ → 3V3 (PU) |

- 1-bit mode uses only DAT0 (DAT1–3 unused). For 4-bit, add DAT1/2/3 with pull-ups.
- Keep CLK trace short; SDMMC runs at MHz clocks. Format both cards FAT32.
- Distribute the Config Card write-protected in production. To reconfigure an aircraft, swap only the Config Card; to update checklists, swap only the Data Card.

### 3.4 Discrete Inputs

| Input                  | ESP32-S3 | Active            | Idle                     |
|------------------------|----------|-------------------|--------------------------|
| SELECT switch (IN/OUT) | GPIO10   | LOW = selected IN | internal PU → HIGH = OUT |
| Push-to-talk           | GPIO0    | LOW = pressed     | internal PU → HIGH       |

Wire each as a simple SPST contact to GND. No external resistor needed (internal pull-ups enabled in firmware). Debounce is handled in software.

## 4 Annunciator Switch & Lamp Driver

Applied Avionics split-legend switch with two independently driven legend halves, following dark-cockpit philosophy per [FAA AC 25-11B](#) and the [AEA design guidance](#).

| State                | TOP — white VOICE CHKLST OFF | BOTTOM — amber VOICE CHKLST FAULT |
|----------------------|------------------------------|-----------------------------------|
| Selected IN, healthy | dark                         | dark ← true dark cockpit          |
| Selected IN, fault   | dark                         | amber ON                          |
| Selected OUT         | white ON                     | dark (fault inhibited)            |

- Power-up lamp test: both halves on for LAMP\_TEST\_MS (~2 s) so a dead LED cannot be mistaken for a healthy dark state, then dark.
- OUT inhibits fault: a deliberately deselected system needs no crew action, so AC 25-11 keeps the amber caution off; only the white OFF status shows. Recognition and fault monitoring are suspended while OUT.

### 4.1 VIVISUN Voltage Options

Applied Avionics VIVISUN / Korrry lighted pushbuttons are offered in 28 VDC, 5 VDC, 28 VAC, 5 VAC, 115 V lamp variants. [Applied Avionics](#). An ESP32 GPIO cannot drive a 28 V (or even 5 V at lamp current) legend directly — use a low-side driver per half (below). For a quick bench mock-up, substitute two ordinary 3.3 V LEDs (white + amber) with series resistors driven straight from GPIO21/GPIO14, observing the ~20 mA GPIO limit.

### 4.2 Lamp-Driver Circuit (one per legend half)

ESP32-S3 GPIO sources ~20 mA default (40 mA max, configurable), with a 1.5 A total chip limit. [ESP32 GPIO current](#). Legend lamps need their own supply and a switch device:

```

+V_lamp (5 V or 28 V, separate rail)
|
[ legend lamp ]      <- VIVISUN legend (white or amber)
|
+----- Drain
GPIO21 --[1k]--|G   N-ch MOSFET (logic-level, e.g. 2N7002 / A03400)
(or G14)      |   or NPN (e.g. 2N2222 with base resistor)
[10k]         <- gate/base pulldown -> GND (defined OFF)
|
GND (Source) ----- common ground with ESP32

```

- One driver for GPIO21 (OFF/white half), one for GPIO14 (FAULT/amber half).
- 10 kΩ gate/base pulldown guarantees the lamp is OFF during boot/reset before the GPIO is configured (dark-cockpit integrity).
- Choose a MOSFET/transistor and supply rated for the lamp. For a 28 V incandescent legend, ensure FET  $V_{ds} \geq 40$  V and current  $\geq$  lamp inrush.
- Tie the lamp supply ground to the ESP32 ground (single common ground).

## 5 Power Budget & Supply

| Rail      | Loads                                                                                                                                                                                   | Typical    | Peak                                                        |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------------------------|
| 3.3 V     | ESP32-S3 + Wi-Fi off + audio-panel input stage (codec) + two microSD                                                                                                                    | ~80–150 mA | ~250 mA (SD init / SR burst)                                |
| 5 V       | Audio output line driver / isolated output stage (installed) — a few mA line-level. <i>Bench only: MAX98357A peaks ~650 mA at 5 V/4 <math>\Omega</math> — not the installed figure.</i> | a few mA   | ~50 mA (installed output stage; $\square$ bench amp figure) |
| Lamp rail | up to 2 legend halves                                                                                                                                                                   | 0 (dark)   | per lamp spec (e.g. 28 V incand.)                           |

- Power from USB 5 V capable of  $\geq 1$  A (the on-board 3.3 V LDO feeds the S3, codec, and SD). The removed MAX98357A was the dominant 5 V load; the installed isolated output stage is much lower current.
- Keep the legend-lamp supply separate from the logic 5 V if using 28 V lamps; only the grounds are common.
- Add bulk decoupling:  $\geq 100$   $\mu$ F near the 5 V rail plus 0.1  $\mu$ F per device. Isolation transformers (input and output paths) are passive.

## 6 Bill of Materials (DIY build)

| Qty            | Part                           | Spec / example                                                                                                                                                | Approx.    |
|----------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 1              | ESP32-S3 DevKit                | DevKitC-1 N16R8 (PSRAM!)                                                                                                                                      | \$12–18    |
| 1              | Audio input stage              | 600 $\Omega$ :600 $\Omega$ aviation audio isolation transformer (e.g. Allen Avionics AGL series) + I2S codec ADC. Selectable analog/digital per install.      | varies     |
| 1              | Audio output stage (installed) | I2S DAC (e.g. PCM5102A-class) + 600 $\Omega$ :600 $\Omega$ line-level isolation transformer $\rightarrow$ COM3-style audio-panel channel. No onboard speaker. | varies     |
| 2              | microSD card + breakout        | FAT32 — Config Card (slot 1, write-protected) + Data Card (slot 2)                                                                                            | \$5–8 each |
| 1              | Annunciator switch             | Applied Avionics VIVISUN/Korry split-legend (or 2 LEDs for bench)                                                                                             | varies     |
| 2              | Lamp driver                    | logic-level N-MOSFET (2N7002/AO3400) or NPN (2N2222)                                                                                                          | <\$1       |
| 4              | Resistors                      | 1 k $\Omega$ $\times$ 2 (gate), 10 k $\Omega$ $\times$ 2 (pulldown)                                                                                           | <\$1       |
| 2–3            | Pull-ups                       | 10 k $\Omega$ on SD CMD/DAT0/CD lines (if breakout lacks them)                                                                                                | <\$1       |
| —              | Caps                           | 0.1 $\mu$ F per device, 100 $\mu$ F bulk near 5 V rail                                                                                                        | <\$1       |
| 1              | PTT button                     | momentary SPST (or use BOOT)                                                                                                                                  | <\$1       |
| *(bench only)* | I2S amp (bench test)           | MAX98357A breakout — bench use only; not installed                                                                                                            | \$4–7      |

| Qty            | Part                  | Spec / example                                                 | Approx. |
|----------------|-----------------------|----------------------------------------------------------------|---------|
| *(bench only)* | Speaker (bench test)  | 4–8 $\Omega$ , $\geq 2$ W — bench use only; not installed      | \$2–5   |
| *(bench only)* | MEMS mic (bench test) | INMP441 or ICS-43434 — bench use only; not the installed input | \$4–6   |

Integrated alternative: ESP32-S3-Korvo-2 (~\$45–55); note its ES8311 codec + NS4150 amp are bench-like configurations. The installed design still requires the isolated audio-panel input stage and isolated line-level output stage (COM3 channel) added externally.

## 7 ESP32-S3-Korvo-2 Differences (integrated build)

- Audio codec is ES8311 (I2C control + I2S data) with an NS4150 speaker amp — not the MAX98357A path. Initialize the codec over I2C (see `esp_codec_dev/BSP`).
- Dual mics feed the AFE — markedly better recognition in noise than one INMP441.
- microSD slot is wired on the board; map `SD_*` pins to the Korvo-2 schematic.
- Expose the annunciator on free header GPIOs; keep the same firmware logic.

## 8 Firmware Build & Behavior

| Topic            | Value                                                                            |
|------------------|----------------------------------------------------------------------------------|
| Framework        | ESP-IDF ≥ 5.2                                                                    |
| Components       | esp-sr, esp_spiffs, driver, json (cJSON), fatfs, sdmmc, esp_driver_sdmmc         |
| Speech models    | WakeNet WN9_HIESP; MultiNet English mn6_en/mn7_en (S3)                           |
| Partitions       | factory app 3 MB + model 5 MB + storage (UI clips) 2 MB → needs 16 MB flash      |
| MultiNet phrases | lowercase letters + single spaces only; spell numbers (“v one”); ~200 cmd cap    |
| Fault behavior   | any SD/parse/validation/audio fault → ST_FAULT, amber legend, no checklist shown |

The partition table assumes a 16 MB flash part (N16R8). An 8 MB module will not fit the 3 MB app + 5 MB model + 2 MB storage layout.

### 8.1 Build & Flash

```
cd firmware
# 1) Render the spoken checklist clips into ./audio (pick your TTS engine):
python3 scripts/make_audio.py --engine say          # macOS
# or --engine espeak or --engine piper --voice
# 2) Target the S3 and build/flash (models + audio embedded automatically):
idf.py set-target esp32s3
idf.py menuconfig # optional - defaults select Hi ESP + mn6_en
idf.py build flash monitor
```

### 8.2 State Machine & Fault Codes

The firmware mirrors the web demo with four states. HOME recognises a checklist trigger and loads it; RUN recognises each item's advance word and steps forward. To keep recognition fast, only the phrases relevant to the current step are registered (esp\_mn\_commands\_add + esp\_mn\_commands\_update).

| State    | Meaning                                                                               |
|----------|---------------------------------------------------------------------------------------|
| ST_OFF   | SELECT switched OUT — white OFF legend lit, voice ignored, fault monitoring suspended |
| ST_FAULT | SD/data/audio problem — amber FAULT legend lit, cause annunciated, retries loading    |
| ST_HOME  | Healthy & idle — listening for a checklist trigger word                               |
| ST_RUN   | Reading a checklist — listening for the active item's advance word                    |

Per the FAA-aligned safety rule, checklist\_store\_load() returns a specific fault code on any problem and leaves the table empty; the device never presents a partial or stale list. The UI/fault clips live in internal SPIFFS (not the card) so it can still speak a fault with no card inserted.

| Fault code    | Trigger                                                      |
|---------------|--------------------------------------------------------------|
| STORE_OK      | SD mounted, JSON parsed + validated, audio present (healthy) |
| FAULT_NO_CARD | SD not detected / mount failed                               |

| Fault code          | Trigger                                           |
|---------------------|---------------------------------------------------|
| FAULT_NO_AIRCRAFT   | no aircraft folder (or config.txt target missing) |
| FAULT_NO_JSON       | checklists.json missing or unreadable             |
| FAULT_PARSE         | JSON malformed                                    |
| FAULT_VALIDATION    | schema / voice-rule violation, empty data, etc.   |
| FAULT_AUDIO_MISSING | a referenced read-aloud clip is absent            |
| FAULT_NO_MEMORY     | allocation failed while loading                   |

## 9 Processor Selection

### 9.1 What this application actually needs

This is not a general dictation device. The workload is narrow and well bounded:

| Requirement               | Implication                                                                                                                               |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Small, fixed vocabulary   | ~13 checklist triggers + a handful of universal advance words = well under the ~200-command MultiNet cap. No large-vocabulary STT needed. |
| Offline, no connectivity  | A safety/training device should not depend on WiFi, BT, or cloud. Connectivity is a liability, not a feature.                             |
| Deterministic + safe      | Must revert to a safe (unopened) state on any fault; single-chip, well-understood toolchain reduces failure surface.                      |
| Dark-cockpit annunciation | A few GPIO + I2S in/out + SD. No high-end peripherals required.                                                                           |
| Low cost, mature tooling  | Demo/training hardware; reproducible with off-the-shelf parts.                                                                            |

The net: command-recognition on a fixed grammar, not open-ended transcription. That keeps an MCU-class part firmly in scope and makes a Linux SBC overkill.

### 9.2 Recommendation

Keep the ESP32-S3 as the baseline; consider the ESP32-P4 only if you want more headroom. ESP-SR (v2.1+) explicitly supports S3 and P4 for English MultiNet, running wake word + AEC/NS + intent on a single chip. The classic ESP32 is no longer supported by current speech algorithms and should be avoided.

- ESP32-S3 (recommended baseline): Xtensa LX7 dual-core @240 MHz with AI vector instructions, 512 KB SRAM, WiFi+BLE, ~\$8–15. Rated the best AI/voice part in the ESP line, most mature tooling, runs mn6\_en/mn7\_en. The firmware, wiring diagram, and BOM are already built around it. [Espressif ESP-SR](#), [espboards.dev](#).
- ESP32-P4 (upgrade path): dual-core RISC-V up to 400 MHz + AI instructions + a 40 MHz low-power core, 768 KB SRAM, ~2.5× the compute of the S3, runs mn7\_en. No WiFi/BT — for a safety device, arguably a plus (removes an attack/distraction surface). Trade-off: needs a companion radio for any connectivity, and slightly less mature tooling. Good if you later add a display or more audio processing. [Espressif ESP32-P4](#), [Elecrow P4 vs S3](#).

For this fixed-vocabulary workload the S3 has ample margin, so the P4 is a “want more headroom / future display” upgrade rather than a necessity.

### 9.3 Alternatives Considered

| Option                                | What it is                                                               | Verdict for this app                                                                                                                    |
|---------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Syntiant NDP120 (Arduino Nicla Voice) | Always-on Neural Decision Processor, ultra-low-power, embedded Cortex-M0 | Excellent for battery always-on wake-word; overkill here since we have panel power and need full command grammar + audio playback + SD. |

| Option                            | What it is                                                      | Verdict for this app                                                                                                                                         |
|-----------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Picovoice Porcupine + Rhino       | Wake-word + speech-to-intent, offline on Arm Cortex-M4          | Clean fit for fixed grammar, but requires a per-deployment AccessKey (license dependency) — undesirable for a self-contained demo.                           |
| Fluent.ai                         | End-to-end speech-to-intent on Cortex-M4 @100 MHz, multilingual | Good for productized multilingual intent; commercial licensing, less open tooling than ESP-SR.                                                               |
| NXP i.MX RT600                    | Cortex-M33 @300 MHz + Cadence HiFi4 DSP @600 MHz, 4.5 MB SRAM   | Strong audio DSP, but more complex board + toolchain than needed for a fixed 13-checklist grammar.                                                           |
| Raspberry Pi (whisper.cpp / Vosk) | Full offline STT on Linux                                       | Real large-vocabulary transcription, but Linux boot, higher power/cost, non-deterministic boot — overkill and less robust for a fixed-grammar safety device. |

Bottom line: the ESP32-S3 remains the right baseline; the ESP32-P4 is the only “strictly better” silicon in the same family and is worth it only if you want extra compute or a display. The dedicated voice chips (Syntiant, Picovoice, Fluent.ai) and the Pi solve problems this app doesn't have.

## 10 microSD Configuration & JSON Schema (Two-Card Model)

Checklist data is split across two cards. The firmware boots the Config Card first (slot 1, /sdcard-config/config.json), then loads the matching aircraft folder from the Data Card (slot 2, /sdcard-data/<aircraft>/). There is no compiled-in fallback — both cards must be present. A missing or unreadable Config Card is FAULT\_NO\_CONFIG; a missing Data Card is FAULT\_NO\_CARD.

```

SL0T 1 (/sdcard-config/) ← Config Card (write-protected per install)
    config.json           installation config: aircraft, audio source, VOX, options

SL0T 2 (/sdcard-data/) ← Data Card (checklist library + audio)
    CJ2/
        checklists.json    the checklist data (schema below)
        audio/
            engine_fire_1.wav one 16-bit/16 kHz/mono WAV per item "clip"
            engine_fire_2.wav
            ready.wav       complete.wav ...

```

### checklists.json schema

```

{
  "aircraft": "CJ2",
  "universal_advance": ["check", "checked", "complete", "next"],
  "checklists": [
    {
      "id": "engine_fire",
      "title": "Engine Fire",
      "triggers": ["engine fire"],
      "items": [
        { "clip": "engine_fire_1",
          "text": "Throttle ... idle",
          "advance": ["idle"] }
      ]
    }
  ]
}

```

Voice-phrase rule (all triggers, advance, universal\_advance): lowercase letters and single spaces only — no digits or punctuation (MultiNet limitation). Spell numbers as words ("v one"). Limits: ≤ MAX\_ITEMS (12) items, ≤ MAX\_TRIGGERS (4) triggers, ≤ MAX\_ADVANCE (4) advance words per item. The loader validates the whole file and verifies every audio clip exists before going live; any violation = FAULT.

## 11 CJ2 Checklist Library

Aircraft CJ2 · 13 checklists. Universal advance words (accepted on any item): check, checked, complete, next.

Numbers spelled as words for MultiNet (e.g. “v one” = V-one). Each item lists its read-aloud text and the spoken word(s) that advance it.

### Engine Fire

Triggers: engine fire · fire warning

| # | Read-aloud item                               | Advance word(s)       |
|---|-----------------------------------------------|-----------------------|
| 1 | Throttle, affected engine, idle.              | idle                  |
| 2 | Engine fire button, lift cover and push.      | push / pushed         |
| 3 | Either illuminated bottle armed button, push. | push / pushed / armed |

### Takeoff Rejected — Below V1

Triggers: takeoff rejected · abort takeoff · speed below v one

| # | Read-aloud item       | Advance word(s)        |
|---|-----------------------|------------------------|
| 1 | Brakes, as required.  | as required / required |
| 2 | Throttles, idle.      | idle                   |
| 3 | Speed brakes, extend. | extend / extended      |

### Takeoff Continued — Above V1

Triggers: takeoff continued · speed above v one

| # | Read-aloud item                                          | Advance word(s)             |
|---|----------------------------------------------------------|-----------------------------|
| 1 | Maintain directional control.                            | control / maintained        |
| 2 | Accelerate to V R.                                       | accelerating / rotate speed |
| 3 | Rotate at V R, climb at V two.                           | rotate / climbing           |
| 4 | Landing gear up, after positive rate of climb.           | up / gear up                |
| 5 | At fifteen hundred feet, retract flaps, then accelerate. | flaps up / accelerating     |

### Engine Failure During Final Approach

Triggers: engine failure final approach · engine failure approach

| # | Read-aloud item                                 | Advance word(s)         |
|---|-------------------------------------------------|-------------------------|
| 1 | Thrust, operating engine, increase as required. | increased / as required |
| 2 | Airspeed, V approach.                           | v approach / set        |
| 3 | Flaps, takeoff and approach.                    | set / flaps set         |

## Emergency Restart — Two Engines

Triggers: emergency restart · engine restart · restart engines

| # | Read-aloud item                                             | Advance word(s) |
|---|-------------------------------------------------------------|-----------------|
| 1 | Ignition switches, both on.                                 | both on / on    |
| 2 | Left and right fuel boost switches, both on.                | both on / on    |
| 3 | Throttles, idle.                                            | idle            |
| 4 | If altitude allows, increase airspeed to two hundred knots. | increased / set |

## Electrical Fire or Smoke

Triggers: electrical fire · electrical smoke · smoke removal · cabin smoke

| # | Read-aloud item                              | Advance word(s)         |
|---|----------------------------------------------|-------------------------|
| 1 | Oxygen masks, don and emergency.             | on / donned / emergency |
| 2 | Microphone select switches, mic oxygen mask. | set / selected          |

## Cabin Altitude

Triggers: cabin altitude · cabin alt · depressurization

| # | Read-aloud item                                              | Advance word(s)          |
|---|--------------------------------------------------------------|--------------------------|
| 1 | Oxygen masks, don and one hundred percent oxygen.            | on / donned              |
| 2 | Microphone select switches, mic oxygen mask.                 | set / selected           |
| 3 | Emergency descent, as required.                              | as required / descending |
| 4 | Passenger oxygen, make sure passengers are receiving oxygen. | confirmed / receiving    |

## Emergency Descent

Triggers: emergency descent · rapid descent

| # | Read-aloud item                                                  | Advance word(s)    |
|---|------------------------------------------------------------------|--------------------|
| 1 | Autopilot trim disconnect button, press and release.             | pressed / released |
| 2 | Throttles, idle.                                                 | idle               |
| 3 | Speed brakes, extend.                                            | extend / extended  |
| 4 | Airplane pitch attitude, approximately twenty degrees nose down. | nose down / set    |

## Battery Overtemperature

Triggers: battery overtemp · battery overtemperature · batt o temp

| # | Read-aloud item            | Advance word(s)    |
|---|----------------------------|--------------------|
| 1 | Volt amp, note.            | noted / note       |
| 2 | Battery switch, emergency. | emergency / set    |
| 3 | Volt amp, note decrease.   | noted / decreasing |

## Autopilot Malfunction

Triggers: autopilot malfunction · autopilot failure

| # | Read-aloud item                                      | Advance word(s)    |
|---|------------------------------------------------------|--------------------|
| 1 | Autopilot trim disconnect button, press and release. | pressed / released |

## Electric Elevator Trim Runaway

Triggers: trim runaway · elevator trim runaway · elevator trim

| # | Read-aloud item                                      | Advance word(s)    |
|---|------------------------------------------------------|--------------------|
| 1 | Autopilot trim disconnect button, press and release. | pressed / released |
| 2 | Throttles, as required.                              | as required / set  |
| 3 | Speed brakes, as required.                           | as required / set  |
| 4 | Manual elevator trim, as required.                   | as required / set  |
| 5 | Pitch trim circuit breaker, left panel, pull.        | pull / pulled      |

## Emergency Evacuation

Triggers: emergency evacuation · evacuate · evacuation

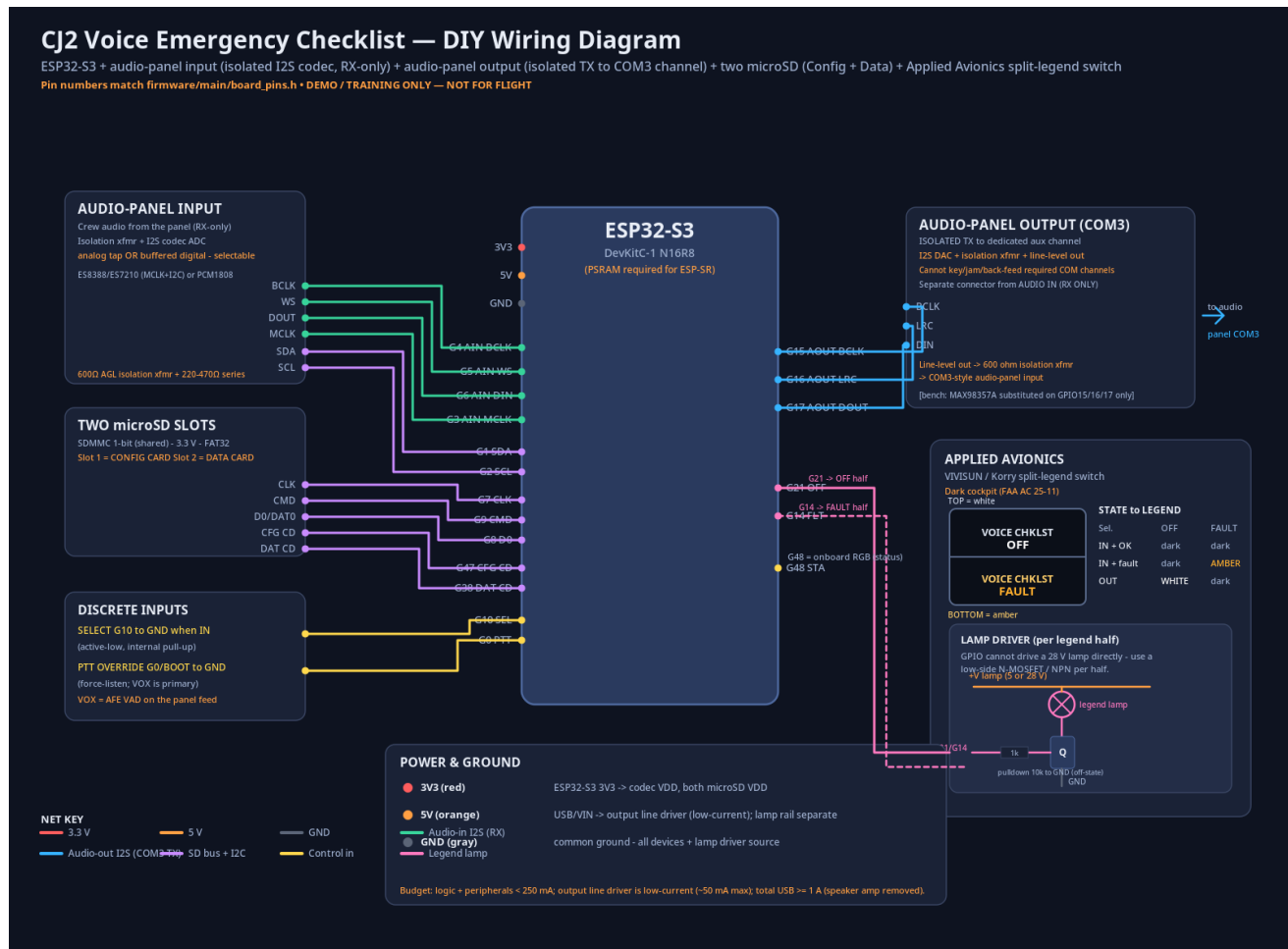
| # | Read-aloud item                                                                                             | Advance word(s)             |
|---|-------------------------------------------------------------------------------------------------------------|-----------------------------|
| 1 | Parking brake, set.                                                                                         | set                         |
| 2 | Throttles, both off.                                                                                        | both off / off              |
| 3 | Left and right engine fire buttons, both press.                                                             | both press / pressed        |
| 4 | Illuminated bottle armed buttons, both press, if fire suspected.                                            | both press / pressed        |
| 5 | Battery switch, off.                                                                                        | off                         |
| 6 | Emergency locator transmitter, make sure the system is activated.                                           | activated / confirmed       |
| 7 | Airplane and immediate area, check for the best escape route.                                               | checked / confirmed         |
| 8 | If exiting through the cabin door, open it. If through the emergency exit, remove and throw the door clear. | cabin door / emergency exit |
| 9 | Move away from the airplane. Procedure complete.                                                            | clear / complete            |

## Before Start

Triggers: before start · before engine start

| # | Read-aloud item                    | Advance word(s)            |
|---|------------------------------------|----------------------------|
| 1 | Parking brake, set.                | set                        |
| 2 | Seat belts, fasten.                | fastened / set             |
| 3 | Battery switch, on.                | on                         |
| 4 | Fuel quantity, check.              | checked / check            |
| 5 | Flight controls, free and correct. | free and correct / checked |

## 12 Wiring Diagram



System wiring diagram: ESP32-S3 + isolated audio-panel input stage (I2S\_NUM\_0 / AIN\_BCLK/LRCLK/DIN, GPIO4/5/6) + isolated audio output stage to COM3-style channel (I2S\_NUM\_1 / AOUT\_BCLK/LRCLK/DOUT, GPIO15/16/17) + two microSD (Config Card + Data Card) + Applied Avionics split-legend annunciator with lamp-driver sub-circuit and dark-cockpit state table. INMP441 mic + MAX98357A bench-test items shown separately.

## 13 Source References

| #  | Reference                                   | Source                                |
|----|---------------------------------------------|---------------------------------------|
| 1  | INMP441 microphone datasheet                | <a href="#">Farnell / InvenSense</a>  |
| 2  | MAX98357A amplifier datasheet               | <a href="#">Analog Devices</a>        |
| 3  | MAX98357A breakout guide                    | <a href="#">Adafruit</a>              |
| 4  | ESP32-S3 GPIO drive current                 | <a href="#">Espressif forum</a>       |
| 5  | ESP32-S3 GPIO reference                     | <a href="#">Espressif docs</a>        |
| 6  | ESP32-S3 reserved/strapping pins (R8 octal) | <a href="#">Arduino forum summary</a> |
| 7  | microSD operating current                   | <a href="#">Arduino forum</a>         |
| 8  | VIVISUN lamp voltage options                | <a href="#">Applied Avionics</a>      |
| 9  | Annunciation color / dark-cockpit guidance  | <a href="#">FAA AC 25-11B</a>         |
| 10 | Avionics display design article             | <a href="#">AEA Avionics News</a>     |
| 11 | ESP-SR speech recognition framework         | <a href="#">Espressif docs</a>        |
| 12 | ESP32 SoC options for AI/voice              | <a href="#">espboards.dev</a>         |
| 13 | ESP32-P4 product page                       | <a href="#">Espressif</a>             |
| 14 | ESP32-P4 vs ESP32-S3 performance            | <a href="#">Elecrow</a>               |
| 15 | Syntiant NDP120 Neural Decision Processor   | <a href="#">Syntiant</a>              |
| 16 | Keyword spotting on microcontrollers        | <a href="#">Picovoice</a>             |

DEMO / TRAINING USE ONLY — NOT FOR ACTUAL FLIGHT OPERATIONS. Checklist text reproduced for demonstration/training only. Repository: [github.com/flas-tech/cj2-voice-emergency-checklist](https://github.com/flas-tech/cj2-voice-emergency-checklist) (MIT License).